

How should a medical student's first scope overview be written?

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Scoping Review is the first step for many junior graduate students or primary care doctors to start academic writing. Compared to systematic reviews (meta-analysis), scope reviews **do not require critical quality evaluation** nor necessarily quantitative integration; their core goal is to "paint a comprehensive picture of research in a particular field and identify knowledge gaps."

Today, I will break down the pure **academic process** and writing principles for writing a high-quality medical overview review, closely following methodological rigor and writing logic throughout.

Step 1: Precisely construct the research question (PCC principle)

The soul of a scope review lies in "defining boundaries," which must strictly use the **PCC** framework, rather than the PICO of clinical research.

- **P (Population):** Identify the key characteristics of the target group (e.g., **elderly patients who have undergone hip fracture surgery**).
- **C (Concept):** Clarifies the core research theme (e.g., **assessment tools and influencing factors for kinesiophobia**). The concept must be clearly defined, not just vague terms like "rehabilitation nursing."
- **C (Context):** Clearly define a specific environment, region, or cultural background (e.g., **primary medical/community rehabilitation institutions in China**).

Writing Conclusion: At the end of the introduction, it is directly stated in a paragraph: "This study raises the following research questions based on the PCC framework: (1) What assessment tools are involved in existing research in this field?" (2) What are the main influencing factors? (3) What knowledge gaps exist in existing literature? "

Step 2: Develop inclusion criteria and search strategies (the foundation of methodology)

(1) Exclusion criteria (a variant of the PICOS principle, but no mandatory control group is required)

It must be clearly stated:

- **Inclusion criteria:**(1) Study subjects meet PCC definitions; (2) Research types include original research (cross-sectional studies, cohorts, RCTs, qualitative studies, **excluding reviews, editorials, and case reports**); (3) Chinese and English literature.
- **Exclusion criteria:**(1) duplicate publication; (2) Unable to obtain the full text; (3) Non-medical or nursing core content.

(2) Retrieval Strategy Construction (Core Operations)

Use a **combination of "subject term + free word."** Take PubMed as an example, and the Chinese repository (CNKI/Wanfang/VIP) works similarly.

- **Step 1:** Accurately check the subject terms in the MeSH database.
- **Step 2:** Based on the key literature found, excerpt high-frequency free words from their titles and abstracts.
- **Step 3:** Write Boolean logic, for example:
`("Diabetes Mellitus"[MeSH] OR "type 2 diabetes") AND ("Wearable Electronic Devices"[MeSH] OR "smartwatch" OR "fitness tracker") AND ("Medication Adherence"[MeSH] OR "compliance" OR "persistence")`

Writing Tips: In the methodology section, at **least one core database must be fully presented with the full search method** (usually PubMed), while the rest of the database must briefly describe the search strategy. Also specify the search period (e.g., database creation up to June 2026).

Step 3: Standardized literature screening (PRISMA flowchart)

This is the **flowchart that a scoping review must present**, and it is also a direct proof of methodological rigor. The process is divided into four levels:

1. **Total number of literatures retrieved from database searches** (n = total of all databases).
2. **After removing duplicate literature** (n = number after duplication, indicate use of EndNote or NoteExpress for duplicate checking).
3. **Initial screening (reading the title and abstract) exclusion** (n = exclusion reasons, such as non-PCC populations, animal experiments, etc.).
4. **Re-screening (read full text) exclusion** (n = exclusion reasons, such as lack of outcome indicators, inconsistent interventions, etc.).
5. **Finally, the literature included** (n = final number).

Key writing points: It must be noted that **"screening is conducted independently by two researchers;** if opinions differ, discussion is held or a decision by a third researcher is required." This step must never be omitted. Even if you do it alone, you should state it in your method to reflect objectivity.

Step 4: Data Extraction (Charting) — Create an "Evidence Summary Table"

Data extraction is not simply copying summaries, but distilling "structured information." You need to design an **Excel extraction table** and present it as a **three-line table** in your thesis.

Core extraction items (mandatory):

First author and year	Country/Region	Types of research design	Sample Characteristics (Sample Size/Age)	Core concepts/interventions	Key outcome measures and key data	Research Conclusions (1 sentence)
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Writing Summary: In the first section of the results section, first describe the basic overview of the included literature in text (annual publication trends, country distribution, proportion of design types), then **directly attach this large table**. The table does not require interpretation, is for readers to consult.

Step 5: Write the "Results" section — the key is "theme clustering"

The result of a scope review is **not a list of literature**, but rather a **categorization and integration**. You need to read through all included literature and divide it into 3-6 subcategories based on **the research topic**.

Common classification dimensions:

- **Dimension One (Current Status):** Current prevalence/occurrence/adherence rate of a phenomenon.
- **Dimension Two (Factor Category):** Influencing factors (demographic factors, disease factors, psychosocial factors).
- **Dimension Three (Tools):** Existing assessment scales/diagnostic tools and their applications.
- **Dimension Four (Intervention Type):** Types and effects of existing intervention strategies.
- **Dimension Five (Experiential Type):** Subjective experiences of patients/healthcare workers revealed through qualitative research.

Writing Skills: Each subcategory has its own section (e.g., "2.1 Current Status Study of XXXX," "2.2 Research on Influencing Factors of XXXX"). Within each section, first state how many studies are included on the topic, then summarize the common findings and controversies of these studies, **supported by specific numbers and scope** (for example: "The 12 included cross-sectional studies show prevalence ranges from 15.3% ~ 42.7%").

Step 6: Write the "Discussion" section — focusing on "knowledge gaps" and "future directions"

The discussion section is not about repeating the results, but about interpreting and elevating them. Discussion of scope reviews follows a fixed three-part logic:

1. **Summary of Main Evidence (Paragraph 1):** Distill and summarize the core areas covered in this review, pointing out which directions are sufficiently evidenced.
2. **In-depth analysis of evidence gaps (main paragraph):** Clearly state why there are gaps in current research. For example:
 - "Currently, cross-sectional studies account for 87%, but **there is a lack of longitudinal tracking studies**, making it impossible to determine causal associations."
 - "Existing interventions are all based on tertiary urban hospitals and **lack contextualized evidence targeting grassroots or remote areas**."
 - "Most studies only measure objective clinical indicators, **lacking patient-reported outcomes (PROs) and quality of life data**."
3. **Methodological Reflection (Paragraph 1):** Honestly discuss the limitations of this review itself (e.g., including only Chinese and English, possibly missing grey literature, not evaluating the quality of included studies, etc.—**note that not conducting quality evaluation is a legitimate feature of scoping reviews, but acknowledge this**)。

Step 7: Refine the conclusion formula

The conclusion should not exceed 200 words and must include the following three elements:

- **Review:** This study systematically depicts the research panorama of the XXX field.
- **Gap:** Currently, high-quality evidence is concentrated in XXX, but lacks XXX.
- **Direction:** In the future, XXX designs (such as multicenter prospective cohorts, randomized controlled trials) should be strengthened or focused on the XXX population.

"Self-check" checklist during the writing process:

- Does the introduction clearly state "why conduct a scope review" (i.e., research in this field is fragmented and lacks systematic integration, rather than simply "lack of research")?
- Does the methodology clearly indicate the **PRISMA-ScR** checklist (not necessarily a full text, but should indicate compliance with the guidelines)?
- Is the search time specified in the "day" (e.g., search deadline: June 1, 2026)?
- Do all charts have independent numbers and headings (Figure 1 flowchart, Table 1 feature table, Figure 2 topic distribution)?
- Were at least three "Knowledge Gaps" clearly and independently listed during the discussion?

- Does the entire text avoid the wording of "quality scoring" or "meta-merging" of included literature (this is the work of systematic reviews; overstepping scope reviews will be rejected)?

Table of Contents Template:

1 Data and Methods

1.1 Establishing Research Questions (PCC Framework)

1.2 Literature Search Strategies

1.3 Inclusion and Exclusion Criteria (According to PICOS Principles)

1.4 Literature Screening and Data Extraction

2 Results

2.1 Literature Screening Process and Basic Features (see Figure 1 and Table 1).

2.2 Distribution Characteristics of Research Topics (see Figure 2)

2.3 Summary

of Existing Research Issues 3 Discussion

(.....)

Recommended popular directions in medical scope overview:

序号	研究方向	核心切入点 (写什么)	预估文献量
1	互联网+护理在慢性伤口居家管理中的风险预警	不写“效果评价”，专写“预警指标”和“家属数字素养”	适中（中文多）
2	机器学习在脓毒症早期预警中的应用	梳理“模型类型”，猛攻“外部验证缺失”这一空白	适中偏多（英文暴涨）
3	GLP-1受体激动剂的胃肠道不良反应管理	避开药理，专写“患者主观体验”与“护理应对策略”	适中（蓝海）
4	老年骨科择期手术的预康复实施范畴	按“运动/营养/心理”三大模块分类归纳	适中（结构最清晰）
5	ICU后综合征家属照护者的心理韧性干预	聚焦“家属端”，总结“干预类型”与“起效时间窗”	适中偏少（交叉热点）

